Assignment 3: Key Concepts and Problem Map

Chapter 3 — determinants

Jingren Wang

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The problem map: 13 questions, four ideas

Concept cluster	Problems	The question it asks	Deciding tool
A. Computing determinants	Q1 Q2 Q4	what is this determinant?	cofactor expansion along sparse rows/columns; permutation $= \pm 1$
B. Determinants as a test	Q3 Q5 Q11	for which parameter does det vanish?	det as a polynomial in the parameter; similar matrices share det
C. Properties of determinants	Q6 Q8 Q9 Q10	what do row/column operations do?	swap flips, scale multiplies, add is free; det not additive
D. Applications	Q7 Q12 Q13	what is det good for?	elimination track (Q7), area (Q12), Cramer's rule (Q13)

- Everything reduces to one skill: reduce a determinant to small ones by expansion along zeros, or track it through row operations.

A · Expansion along zeros does the work (Q1, Q2)

Q1 — expand along the last row

$\begin{bmatrix} 4 & -1 & -4 & 5 \\ 5 & -3 & 4 & -5 \\ 3 & 0 & 0 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$: row 4 has three zeros, so

$$\det = 1 \cdot \det \begin{bmatrix} 4 & -1 & -4 \\ 5 & -3 & 4 \\ 3 & 0 & 0 \end{bmatrix} = 3 \cdot \det \begin{bmatrix} -1 & -4 \\ -3 & 4 \end{bmatrix}$$

$$= 3\,(-4 - 12) = \boxed{-48}$$

Q2 — expand along the LAST column: three zeros

Column 4 is $(4, 0, 0, 0)$ — three zeros, so only ONE term survives:

$$\det = 4 \cdot \underbrace{(-1)^{1+4}}_{-} \det \begin{bmatrix} 4 & -2 & 2 \\ 5 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix} = -4 \det \begin{bmatrix} 4 & -2 & 2 \\ 5 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix}$$

Inside the 3×3, row $2 = (5, 0, 0)$ — expand along it:

$$\det \begin{bmatrix} 4 & -2 & 2 \\ 5 & 0 & 0 \\ -2 & 0 & 4 \end{bmatrix} = 5 \cdot \underbrace{(-1)^{2+1}}_{-} \det \begin{bmatrix} -2 & 2 \\ 0 & 4 \end{bmatrix} = -5\,(-8) = 40$$

so

$$\det = -4 \times 40 = \boxed{-160}$$

- Always expand along the row or column with the most zeros — each zero deletes one 3×3 minor.

A · A permutation matrix, and singular means det = 0 (Q3, Q4)

Q4 — permutation matrix

$P = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ = identity with rows 1 and 3 swapped: one interchange

$\Rightarrow \det P = (-1)^1 \cdot 1 = \boxed{-1}$. (Permutations: $+1$ even, -1 odd.)

Q3 — singular means det = 0

$M = \begin{bmatrix} -2 & -4 & -1 \\ 0 & 1 & 1 \\ k-6 & 10 & 1 \end{bmatrix}$: expand along column 1 —

$\det M = -2(1-10)+(k-6)(-4+1) = 36-3k$

$\det = 0 \Rightarrow \boxed{k = 12}$.

- Q3's det is *linear* in k because column 1 has a zero entry: one term drops out.

B · Determinants as polynomials in a parameter (Q5, Q11)

Q5 — $|A| = 0$ as a quadratic in a

$A = \begin{bmatrix} a & 1 & 1 \\ a & -8 & 2 \\ 9 & -6 & a \end{bmatrix}$, expand along column 1:

$$\det A = a(-8a + 12) - a(a + 6) + 9(10) = -9a^2 + 6a + 90$$

$$|A| = 0 \iff 3a^2 - 2a - 30 = 0 \iff a = \frac{1 \pm \sqrt{91}}{3}.$$

Q11 — similar matrices share det

$\det B = \det(S^{-1}AS) = \det A$, so
$\det B = 0 \iff \det A = 0$. A is block triangular:

$$\det A = (2 - x) \cdot \det \begin{bmatrix} 1-x & -4 \\ 1 & 5-x \end{bmatrix}$$

$$= (2 - x)\left((1 - x)(5 - x) + 4\right) = (2 - x)(x - 3)^2$$

$\Rightarrow x = \mathbf{2}, \mathbf{3}$ (3 is a double root — watch the -4: the 2×2 block is *not* diagonal).

C · Properties at work (Q6, Q8)

Q6 — every entry negated

Negating a whole matrix = scaling each of its 3 rows by -1:

$$\text{det multiplied by } (-1)^3 = -1 \quad \Longrightarrow \quad a = \boxed{-1}$$

No evaluation needed — that is the point of the question.

Q8 — one row op, one column scaling

Start from $\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = -5$:

$$R_1 \leftarrow R_1 - 9R_3 \text{ (free)} : \quad \begin{bmatrix} a-9g & b-9h & c-9i \\ d & e & f \\ g & h & i \end{bmatrix}$$

then scale column 2 by 2: $\det \to 2 \times (-5) = -10$

$\Rightarrow$ target determinant $= 2(-5) = \boxed{-10}$.

C · True/false statements (Q9, Q10)

Q9 — true: A, C, D

A row addition is free ✓; **B** false: $\det(2I) = 2^n$ but $\det I + \det I = 2$; **C** dependent columns $\iff$ singular $\iff$ $\det = 0$ ✓; **D** $\det A = (-1)^r \times$ product of the pivots of any echelon U ✓

Q10 — true: C only

A false: diagonal product only for triangular A; **B** false: $\det\left[\begin{smallmatrix} 1 & 2 \\ 2 & 4 \end{smallmatrix}\right] = 0$ with no equal/zero rows; **C** two swaps $= (-1)^2$: det restored ✓; **D** false: $\det A^\top = \det A$.

- The recurring trap: det is *multiplicative* ($\det AB = \det A \det B$) but never *additive*.

D · Elimination tracks det (Q7)

The setup

$E_4E_3E_2E_1A = U$ with E_1 row addition ($\det 1$), E_2 swap ($\det -1$), E_3 scale R2 by 9 ($\det 9$), E_4 scale R3 by -5 ($\det -5$):

$$\det U = \underbrace{1 \cdot (-1) \cdot 9 \cdot (-5)}_{45} \det A \qquad \Longrightarrow \qquad \det A = \frac{\det U}{45}.$$

$U = \begin{bmatrix} 1 & -9 & 6 \\ 0 & 7 & -3 \\ 0 & 0 & 3 \end{bmatrix}$

U is triangular: $\det U = 1 \cdot 7 \cdot 3 = 21$ (Thm 3.1), so

$$\det A = \frac{\det U}{45} = \frac{21}{45} = \boxed{\tfrac{7}{15}} \qquad (\det A \neq 0 : \ A \text{ is invertible})$$

D · Area by determinant, and Cramer's rule (Q12, Q13)

Q12 — triangle area

Vertices $(3,-3),(12,-6),(5,1)$: translate, then

$$\text{Area} = \tfrac{1}{2}\left|\det\begin{bmatrix} 9 & 2 \\ -3 & 4 \end{bmatrix}\right| = \tfrac{1}{2}|36+6| = \boxed{21}$$

Q13 — Cramer's rule

(a) $\det A = 15 - 150 + 120 = \boxed{-15}$ for $A = \begin{bmatrix} -1 & -2 & -8 \\ -2 & 1 & 7 \\ 3 & 6 & 27 \end{bmatrix}$.

(b) $\det A_i / \det A$:

$$\det A_1, \det A_2, \det A_3 = 27,\ -251,\ 50$$

$$\mathbf{x} = \left(\tfrac{27}{-15},\ \tfrac{-251}{-15},\ \tfrac{50}{-15}\right) = \left(-\tfrac{9}{5},\ \tfrac{251}{15},\ -\tfrac{10}{3}\right)$$

all three equations verified ✓

D · Q13 in detail: Cramer's rule computation

Step 1 — det A, cofactor expansion along row 1

Minors (delete row 1 and the entry's column): $M_{11} = -15$, $M_{12} = -75$, $M_{13} = -15$; cofactors -15, 75, -15 (signs $+, -, +$):

$$\det A = (-1)(-15) + (-2)(75) + (-8)(-15) = 15 - 150 + 120 = \boxed{-15}$$

Step 2 — the three Cramer determinants (column i of A replaced by $\mathbf{b}$)

$$\det A_1 = (-5)(-15) + (-3)(6) + (5)(-6) = 75 - 18 - 30 = \boxed{27}$$

$$\det A_2 = (-5)(75) + (-3)(-3) + (5)(23) = -375 + 9 + 115 = \boxed{-251}$$

$$\det A_3 = (-1)(23) + (-2)(1) + (-5)(-15) = -23 - 2 + 75 = \boxed{50}$$

Step 3 — assemble $\mathbf{x} = (\det A_1/\det A,\ \det A_2/\det A,\ \det A_3/\det A)$: $\mathbf{x} = (\frac{27}{-15},\ \frac{-251}{-15},\ \frac{50}{-15}) = (-\frac{9}{5},\ \frac{251}{15},\ -\frac{10}{3})$ — all three equations verified ✓

One table: the whole assignment

Idea	The one thing to remember
Expansion	expand along zeros (Q1: -48; Q2: -160); permutation $= \pm 1$ (Q4: -1)
Parameter test	det as a polynomial: $36 - 3k$ (Q3: $k = 12$); $-9a^2 + 6a + 90$ (Q5); $(2 - x)(x - 3)^2$ via similarity (Q11: $x = 2, 3$)
Properties	negate all = three row scalings (Q6: $a = -1$); one row op free + column scaling $\times 2$ (Q8: -10); Q9: A, C, D; Q10: C only
Elimination	$\det A = \det U / 45 = 21/45 = 7/15$ (Q7)
Applications	area $= \frac{1}{2}\lvert\det\rvert$ (Q12: 21); Cramer with $\det A = -15$ (Q13: $(-\frac{9}{5}, \frac{251}{15}, -\frac{10}{3})$)

Review map — back to the class slides

Questions	Topic to review	Class slides (file · pages)
Q1 Q2 Q12	cofactor expansion; determinants as area	`CH3.1-3.3.pdf`: pp. 4–8, 25, 32–33
Q3 Q5 Q11	determinants as a test for singularity / parameters	`CH3.1-3.3.pdf`: pp. 13, 16 (properties), 9 (triangular)
Q4 Q9 Q10	row operations and permutation structure	`CH3.1-3.3.pdf`: pp. 11–12
Q7	elimination track: pivots and swaps	`CH3.1-3.3.pdf`: pp. 9, 11
Q8	column operations and multilinearity	`CH3.1-3.3.pdf`: pp. 14–16
Q13	Cramer's rule	`CH3.1-3.3.pdf`: pp. 17–20

All files live in `Lecture slides/chapter 3/`.

Answer key (instructor copy, reference only)

- Q1: -48
- Q2: -160
- Q3: $k = 12$
- Q4: $\det P = -1$
- Q5: $a = (1 \pm \sqrt{91})/3$
- Q6: $a = -1$
- Q7: $\det A = \det U/45 = 21/45 = 7/15$ ($U = [[1, -9, 6], [0, 7, -3], [0, 0, 3]]$)
- Q8: -10
- Q9: A, C, D
- Q10: C only
- Q11: $x = 2, 3$
- Q12: Area $= 21$
- Q13: (a) $\det A = -15$; (b) $x_1 = -9/5$, $x_2 = 251/15$, $x_3 = -10/3$

All answers verified by exact-arithmetic scripts (`chapter3_solutions/`).

Thank you!

Questions? Comments? Corrections?

Write to me: jwang929@connect.hkust-gz.edu.cn